Majid Farhadi

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HIGHLIGHTS

- Extensive hands-on experience in digital design, computer architecture, FPGA synthesis and implementation, **RTL** design, high-speed **AXI** interconnects, resource and **power** optimization, clock and **timing** management, static timing analysis (**STA**), simulation, waveform analysis, **verification**, and testing.
- Extensive experience in **embedded** system, **software**, and **firmware** design and **debugging**, SoC and HW/SW co-design, edge IoT applications ranging from **bare-metal** to **kernel-based** systems, UART/SPI/I2C and TCP/IP communication protocols, ADC/DAC peripherals, and AVR/ARM/ESP/STM32 micro-controllers.
- 5 years of research in AI/ML, collaborative **teamwork**, cross-functional communication, and documentation.
- 7 years of **teaching** assistantship and mentorship experience across **20+** undergraduate and graduate courses, leading laboratory sessions and supporting lecture classes ranging from 10 to over 100 students.

TECHNICAL SKILLS

Programming Languages: Verilog, VHDL, C, C++, Python, Bash, TCL, CMake, Assembly, MATLAB

Tools and Frameworks: Xilinx Vivado, Quartus Prime, ModelSim, Xilinx ISE, Platform Designer, Altium Designer, STM32CubeMX, Arduino IDE, Atmel Studio, Raspberry Pi Pico SDK, GDB, FreeRTOS, Git

EDUCATION

Master of Science in Computer Engineering (GPA: 3.9/4) **Sep 2023 – Mar 2026**
Department of Electrical and Computer Engineering, University of Alberta Edmonton, Canada

Bachelor of Science in Computer Engineering (GPA: 18.36/20 (3.9/4)) **Sep 2018 – Feb 2023**
Department of Electrical and Computer Engineering, Isfahan University of Technology Isfahan, Iran

RESEARCH AND WORK EXPERIENCES

Graduate Research Assistant **Sep 2023 – Present**
Very Large Scale Integration (VLSI) Research Lab, University of Alberta

- Designed and implemented **AI architectures**, including training and inference circuits, on **FPGAs**.
- Integrated and **synchronized IP** core modules with **edge detector** 1-stage synchronizers.
- Wrote **testbench** modules for behavioral, functional, and timing **simulation** and verification of **RTL** designs.
- Defined implementation **constraints** including **clock** period to meet **timing** requirements.
- Optimized **dynamic power** by minimizing **switching** activity through **stop** conditions.
- Fine-tuned distance functions in clustering algorithms and derived gradients in backpropagation algorithms.

Graduate Research Assistant **Jan 2023 – May 2024**
Structured Machine Learning Research Lab, Simon Fraser University

- Counted **motifs** in a **graph database** by **matrix multiplication**, suitable for ML applications.

Undergraduate Research Assistant **Sep 2021 – Feb 2023**
Embedded Systems and Networks Research Lab, Isfahan University of Technology

- Designed an **interface** for a hardware accelerator for the time-slotted channel hopping (TSCH) protocol.
- Stacked on top of an IEEE 802.15.4 transceiver used in wireless sensor networks in **IoT** applications.
- Deployed on an Altera DE1-**SoC** board, featuring a Cyclone FPGA and a dual-core Cortex-A9 ARM.
- Integrated the ARM processors and FPGA fabric through a lightweight master-slave **AXI** bridge.
- Enabled full-duplex **read/write** communication between FPGA and ARM processor.
- Compiled C programs for the target ARM architecture using **makefiles**.
- Accessed the FPGA from C programs via virtual addresses provided by **memory mappings**.
- Offloaded computation from the processor and exploited the FPGA's high **throughput**.

Intern **Jul 2021 – Sep 2021**
Kian-Pardaz Naghsh-e-Jahan Engineering Company

- Designed a **PCB** and implemented a **web client** for a smart thermostat on an ESP8266-12E module.

PROJECTS

- Soft-Core Implementation of MIPS Architecture on FPGA.
- Adding System Call to XV6 Kernel.
- Development of Smart Health Bracelet to Communicate Vital Signs by Arduino Board and GSM Module.
- ECG Signals Analysis on IoT Platform by Autoencoders.
- Prediction of DDoS Botnet Attacks on IoT Devices by Data Mining Methods.
- Implementation of Autonomous Obstacle Avoiding Robot by Arduino Board and Ultrasonic Sensor.
- Image Seam Carving based on Depth Map.
- Time-Series Forecasting with Deep and Recurrent Neural Networks.

PUBLICATIONS

[1] **M. Farhadi**, W. Pedrycz, and J. Han, "SCFNNs: Stochastic Computing Fuzzy Neural Networks," IEEE Embedded Systems Letters, 2026.

[2] A. Khazraee, A. Mirzaei, **M. Farhadi**, P. Nadaf, K. Zahirnia, M. Salameh, K. Cannons, R. Mar, M. Wu, and O. Schulte, "Rule-enhanced graph learning," in ICML 2024 Workshop on Structured Probabilistic Inference & Generative Modeling, 2024.

[3] **M. Farhadi**, W. Pedrycz, and J. Han, "1BCMs: 1-Bit Cognitive Maps," under review at IEEE Design & Test.

[4] W. Lu, **M. Farhadi**, X. Gao, C. Lynch, and J. Han, "A Memory-Efficient and Low-Complexity Probabilistic Sigmoidal Approximator," under review at Design, Automation and Test in Europe (DATE) Conference.

[5] A. Khazraee, A. Mirzaei, **M. Farhadi**, P. Nadaf, K. Zahirnia, M. Salameh, K. Cannons, R. Mar, M. Wu, and O. Schulte, "Enhancing Graph Generation With First-Order Logic Rules." under reivew at ICLR.

VOLUNTEER SERVICES

Sub-reviewer for Conference and Journal Papers **Sep 2023 – Mar 2026**

• DDECS 2026 • ICLR 2026 • DATE 2026 • ASPDAC 2026 • IEEE TVLSI • NeurIPS 2025 • MLCAD 2025 • ICCAD 2025 • IEEE TETC • LAD 2025 • IEEE ESL • DATE 2025 • ASPDAC 2025 • MLCAD 2024 • DFT 2024 • ICCAD 2024 • DDECS 2024 • ISCAS 2024 • VLSID 2024

Mentoring Undergraduate Students for Dean's Research Awards Program **Sep 2024 – Dec 2024**

HONORS

- Ranked **8th** in the final stage of the national scientific Olympiad of Computer Engineering in 2022.
- Ranked **8th** in Computer Engineering at the Isfahan University of Technology among more than 90 students.
- Ranked **13th** in the national entrance exam for Computer Engineering graduate program among more than 11000 applicants in 2022.
- Ranked **17th** in the national entrance exam for Computer Engineering graduate program among more than 10000 applicants in 2023.

TEACHING EXPERIENCE

Graduate Teaching Assistant **Jan 2024 – Apr 2026**

Department of Electrical and Computer Engineering, University of Alberta

- Introduction to Engineering Design and Communication
- Discrete Time Signals and Systems, Fall 2025
- Embedded System Design, Fall 2024, Fall 2025
- Special Topics in Computer Engineering, 2026
- Computer Interfacing, Winter 2025
- Continuous Time Signals and Systems, Winter 2024

Undergraduate Teaching Assistant **Sep 2019 – Jun 2023**

Department of Electrical and Computer Engineering, Isfahan University of Technology

- Algorithms Design, Spring 2023
- Embedded Systems, Fall 2022
- Robotics, Spring 2022
- Artificial Intelligence, Spring 2022
- Microprocessors, Fall 2021
- Hardware Description Languages, Spring 2021, Fall 2020
- Advanced Programming, Spring 2023 & 2020, Fall 2019
- Formal Languages and Automata Theory, Fall 2022
- Computational Intelligence, Spring 2022
- Operating Systems, Fall 2021
- Computer Architecture, Spring 2021
- Data Structures, Spring 2020